

EN401 — Handout 5 Results & Analysis Report

Commercial Thermal Generation Era & 99.96% Cost Collapse

1. Solver Optimization Metrics

- **Status:** OPTIMAL | **Objective Value (NPV):** \$19,335.91 (-99.96% drop vs. HO4)
- **Problem Scale:** 1,502 Rows, 960 Columns, 8,640 Non-Zeros | **Runtime:** < 0.08s

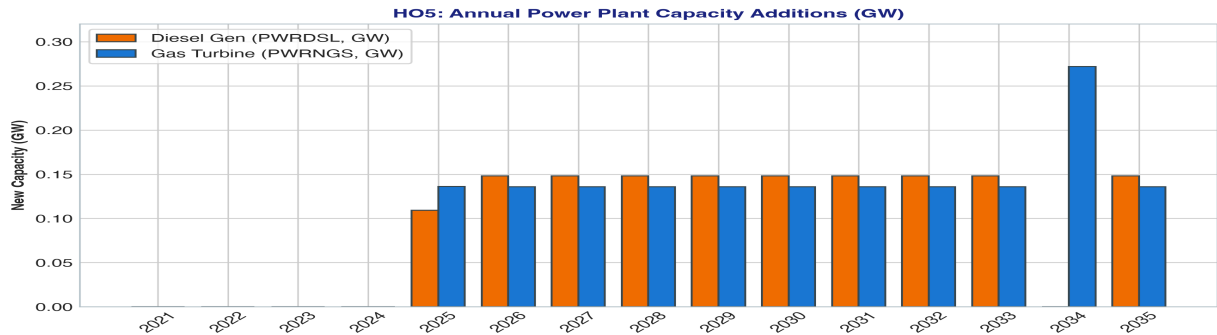


Figure: Gas turbine and diesel generator additions starting in 2025.

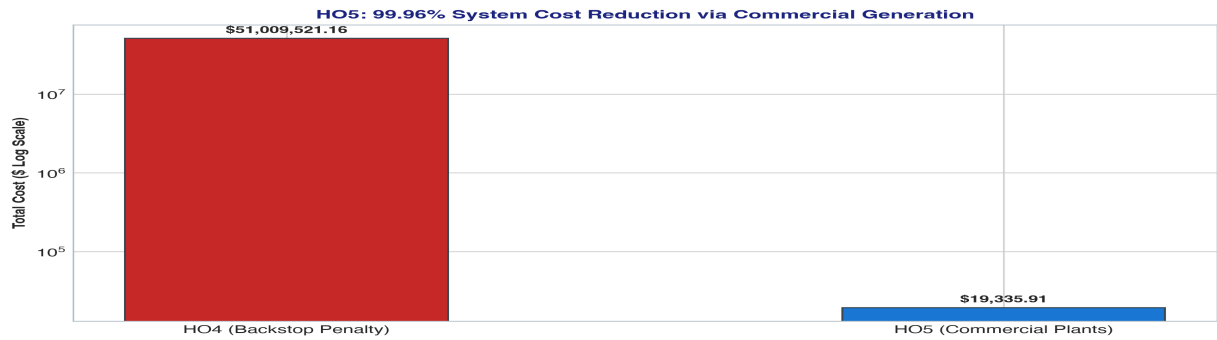


Figure: 99.96% system cost collapse from \$51.01M to \$19.34k.

2. Generation Capacity & Fuel Supply Table

Metric	2021	2023	2025	2027	2029	2031	2033	2035
Gas Mining (PJ)	53.7	6.9	7.6	15.2	7.6	7.6	7.6	7.6
Diesel Import (PJ)	10.6	11.6	21.4	10.7	10.7	10.7	10.7	10.7
New Gas Turb (GW)	0.000	0.000	0.136	0.136	0.136	0.136	0.136	0.136
New Diesel Gen (GW)	0.000	0.000	0.109	0.148	0.148	0.148	0.148	0.148

3. Economic Takeaway

Gas turbines provide baseload energy, while diesel generators cover flexible dry-season peaking.